Temporal and Evidence-Source Provenance-Aware Multimodal Misinformation Detection with Calibrated Abstention

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***Abstract*—Multimodal misinformation detectors can retrieve convincing but temporally invalid, duplicated, or mutually dependent evidence. This paper specifies TPG-Fact, a temporal and evidence-source provenance-aware framework for image–text claim verification. The framework retrieves only evidence available before a declared decision cutoff, groups copied and syndicated material into dependency clusters, represents cross-document and cross-modal relationships in a heterogeneous graph, and separates evidence sufficiency from veracity prediction. The intended output is supported, refuted, or insufficient evidence, with validation-fitted calibration and selective abstention. We also define US-TPMFC, a proposed temporally partitioned benchmark of United States events derived from professional fact-check metadata and open web archives. The evaluation protocol uses future-event splits, retrieval-quality measures, event-level uncertainty, temporal-leakage tests, and controlled duplicate-source interventions. This document describes the preregistered design and executable research scaffold; real-corpus results are intentionally withheld until the planned data audit and experiments are completed.**

***Index Terms*—multimodal misinformation, automated fact-checking, evidence provenance, temporal retrieval, selective prediction, calibrated abstention**

Research status: The method and evaluation protocol are implemented as a reproducible scaffold. No real-corpus or comparative findings are claimed in this version.

I. Introduction

Images are frequently reused with misleading captions, dates, locations, or event descriptions. Because the pixels may be authentic, detectors that emphasize manipulation artifacts or surface-level image–text compatibility can miss the misinformation mechanism. Automated fact-checking therefore increasingly uses external evidence, but retrieved documents create new failure modes: evidence may have been published after the required decision time, multiple results may copy the same origin, and a fluent model may issue a confident verdict when the retrieved material is insufficient.

Knowledge-aware and evidence-grounded multimodal systems address parts of this problem. KAMP learns from text, images, and background knowledge graphs [1]; RED-DOT explicitly reranks relevant multimodal evidence [2]; COVE predicts an image's original context before caption veracity [3]; and multimodal RAG systems retrieve web information for claim reasoning [4], [5]. These advances do not by themselves guarantee that evidence was available at decision time or that apparently diverse results are independent.

We study the following question: *Does reasoning over time-valid, independently sourced evidence improve future-event multimodal misinformation detection and reduce overconfident decisions compared with ordinary multimodal RAG?* The intended operating unit is an English image–text claim about a United States event. The system returns supported, refuted, or insufficient evidence and records the evidence and cutoff used for that decision.

This work makes four planned contributions. First, it defines US-TPMFC, a temporally partitioned benchmark constructed from professional fact-check metadata and open archives. Second, it introduces a temporal-provenance graph that represents publication order, duplication, hyperlinks, publishers, entities, events, and shared media. Third, it proposes a hierarchical detector that assesses evidence sufficiency before veracity. Fourth, it preregisters future-event, leakage, duplicate-source, calibration, and selective-risk evaluations. These contributions remain proposed until the corresponding real-data experiments pass the stated checks.

II. Related Work

A. Multimodal Misinformation Detection

NewsCLIPpings established a large benchmark of authentic but deliberately mismatched news images and captions [6]. VERITE subsequently controlled unimodal bias and topic imbalance in real-world multimodal misinformation evaluation [7]. KAMP introduced knowledge-aware multimodal pretraining using single-modality, cross-modal, and knowledge-content objectives [1]. Synthetic multimodal examples have also been used to improve transfer to real misinformation cases [8]. Event-Radar models visual manipulation, textual emotion, and event-level inconsistency through an event graph [9]. These

content-focused approaches motivate strong no-retrieval and event-aware baselines.

B. Evidence-Grounded Fact-Checking

MOCHEG formulates multimodal fact-checking as evidence retrieval, claim verification, and explanation generation [10]. MetaSumPerceiver summarizes evidence across multiple multimodal documents [11]. RED-DOT challenges the assumption that all web evidence is relevant and introduces relevance-aware reranking [2]. RAGAR and MultiReflect combine multimodal large language models with retrieval and iterative reasoning [4], [5]. SNIFFER uses external retrieval for out-of-context detection [12]. Consequently, retrieval itself is not the novelty claimed here; the proposed distinction is the joint enforcement of temporal validity and source-dependency-aware aggregation.

C. Image Context, Time, and Evidence Dependence

The 5Pils work recovers original image meta-context from web evidence [13], while COVE predicts image context before assessing a caption [3]. Temporal evaluation is essential because multimodal misinformation models lose performance under chronological distribution shift [14]. ChronoFact demonstrates structured timeline reasoning for complex temporal claims [15]. Research on conflicting evidence further shows that source information affects RAG-based fact verification [16]. The present study connects these concerns by representing time and content dependence within the same evidence graph.

III. Task and Benchmark Construction

A. Task Definition

Each instance consists of claim text x_t, an associated image x_v, a decision cutoff τ, and an evidence collection $E_{\leq\tau}$. The output label is $y \in \{\text{supported}, \text{refuted}, \text{insufficient}\}$. Evidence published after τ is unavailable by definition. The independent unit for splitting and uncertainty analysis is the event, not the individual post or document.

B. Candidate Acquisition

The proposed corpus uses English ClaimReview records concerning United States events from 2020–2026. GDELT supplies reproducible discovery metadata and Common Crawl supplies archived page observations. The collection retains cases with an identifiable claim time, recoverable image or archive, unambiguous professional rating, and sufficient timestamp metadata. Fact-check domains and documents containing the final verdict are excluded from the retrieval pool.

C. Label Policy

Only explicit site ratings are normalized. True or correct maps to supported; false, fake, or miscaptioned maps to refuted; and unproven, unsubstantiated, or no evidence maps to insufficient evidence. Mixed, partially true, satire, opinion, and ambiguous ratings are excluded. The project does not introduce new human labels. Automatically extracted evidence links are treated as silver evidence.

D. Temporal Partitioning and Release

Training covers 2020–2023, validation uses 2024, and final testing uses 2025–2026. Event identifiers, near-duplicate claims, and identical perceptual image hashes are group constraints. The proposed release contains URLs, cryptographic and perceptual hashes, timestamps, normalized labels, graph annotations, queries, and split manifests without redistributing copyrighted content.

Before treating US-TPMFC as a benchmark contribution, the filtered corpus must contain at least 2,000 cases, 300 per label, 50 test events, and two independent evidence clusters for at least 70% of retained claims. Otherwise, it will be reported only as a stress-test collection.

IV. Temporal-Provenance Method

A. Time-Constrained Multimodal Retrieval

The claim representation combines named entities, temporal expressions, OCR text, and an image caption. Text candidates are retrieved using BM25 and BGE-M3; visual candidates use SigLIP2 embeddings and perceptual hashes. Reciprocal-rank fusion combines lexical, dense, and visual rankings. The top 100 candidates are reranked, after which the top 20 satisfying the publication-time cutoff are retained.

B. Evidence-Source Provenance Graph

The graph contains claim, image, document, publisher, entity, and event nodes. Typed edges represent retrieval relevance, image containment, near-duplicate text or images, hyperlinks, publisher membership, entity or event mentions, and temporal precedence. Document clusters are estimated using exact hashes, MinHash-style lexical similarity, semantic similarity, perceptual image distance, hyperlinks, and publication order. Because true authorship may be unknowable, the earliest observed record is described only as an origin candidate.

Let $C(E)$ denote the partition of evidence documents into dependency clusters. Instead of summing document-level support, the model first aggregates within each cluster and then combines clusters. This prevents repeated copies of one source from receiving the same weight as multiple independent sources.

C. Hierarchical Decision and Abstention

A heterogeneous graph transformer encodes the temporal-provenance graph conditioned on the claim. A sufficiency head estimates $p(s = 1 \mid x, E_{\leq\tau})$. If evidence is sufficient, a veracity head estimates supported versus refuted. Otherwise, the system returns insufficient evidence. Temperature scaling is fitted on validation predictions, and a validation-selected threshold targets 80%

answer coverage. The final prediction record includes class probabilities, evidence identifiers, source clusters, and the enforced cutoff.

V. Experimental Protocol

A. Baselines

Controlled baselines include majority prediction, text-only classification, image–text fusion without retrieval, an open multimodal language model under zero- and few-shot prompting, vanilla multimodal RAG, time-filtered RAG without provenance, provenance-aware RAG without time filtering, RED-DOT where protocol compatibility permits, and the complete model. Every controlled method uses the same splits, candidate corpus, evaluation code, and 20-trial validation budget.

B. Metrics and Statistical Analysis

The primary metric is macro-F1 on the 2025–2026 event-disjoint test set. Secondary metrics are per-class F1, balanced accuracy, macro one-versus-rest AUROC, Brier score, negative log-likelihood, expected calibration error, area under the risk–coverage curve, risk at 80% coverage, Recall@10, nDCG@10, latency, GPU memory, and retrieved-document count. Final configurations are evaluated with five seeds. Differences use 10,000 paired event-level bootstrap replicates, with Holm correction across the primary comparison family.

C. Ablation and Stress Tests

The preregistered ablations remove temporal filtering, dependency clustering, temporal edge features, graph aggregation, text evidence, visual evidence, the sufficiency gate, and evidence-dropout training. Robustness tests inject syndicated copies, add post-cutoff verdict-bearing documents, remove the earliest observed source, introduce dependent and independent conflicts, swap captions, and reuse historical images.

D. Selection and Falsification Rules

Hyperparameters and the abstention threshold are selected on validation data only. The provenance-performance claim is removed if the component improves neither natural future-event performance nor duplicate-injection robustness. The abstention claim is removed if selective risk fails to improve at matched coverage. Any advantage arising from post-cutoff evidence is labeled leakage rather than superiority.

VI. Results

Research status: Real-data collection, model training, and comparative evaluation have not been executed. This section therefore contains no scientific performance claims.

The repository includes a synthetic smoke test that exercises schema validation, temporal filtering, retrieval, provenance clustering, hierarchical decision output, and metric generation. Its values are software diagnostics and are not comparable with published benchmarks.

TABLE I
Synthetic software smoke-test diagnostics. These values are not scientific results.

Diagnostic	Value
Records	6
Macro-F1	1.000
Balanced accuracy	1.000
Brier score	0.060
ECE	0.200

The final results section will present, in order, corpus statistics, evidence-retrieval results, the controlled main comparison, calibration and selective-risk findings, ablations, temporal and duplicate-source stress tests, and external-dataset checks. Every populated table will be generated from preserved raw predictions.

VII. Discussion, Ethics, and Limitations

The principal methodological risk is mistaking accessible web redundancy for independent corroboration. The proposed dependency clusters are estimates based on observable content and links; they do not prove journalistic independence or original authorship. Similarly, archived timestamps indicate observed availability and may differ from the true first publication time.

The corpus inherits professional fact-check labels and their editorial definitions. Restriction to explicit rating mappings reduces ambiguity but can create selection bias and class imbalance. United States events and English sources limit geographic and linguistic generalization. Common Crawl and GDELT coverage is incomplete, and removed or blocked content may be systematically different from recoverable content.

No copyrighted full text or images will be redistributed without permission. The release will emphasize URLs, hashes, derived metadata, split manifests, and collection code. The system is intended to support research and human review, not to serve as an autonomous censorship or content-removal mechanism. A prediction of insufficient evidence should trigger further investigation rather than imply that a claim is false.

VIII. Conclusion

This paper defines a reproducible study of temporal and evidence-source provenance in multimodal misinformation detection. The central design combines time-constrained retrieval, dependency-aware graph aggregation, evidence-sufficiency prediction, and calibrated abstention. Its evaluation prioritizes future-event generalization, temporal leakage, duplicate-source robustness, and selective risk. The current artifact establishes the protocol and executable scaffold without presenting unmeasured findings. Completion requires freezing the real corpus, running controlled baselines and five-seed experiments, auditing all outputs, and narrowing the final claims to the observed evidence.

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